

## Data Analytics Capstone

### Air Quality Risk Prediction Using Environmental and Meteorological Data

Interim Report

Klarí van Zyl

Walsh College

QM640: Data Analytics Capstone

Mr Abhay Poddar

Fall 2025 Term 3

6 May 2026

The GitHub repository for this capstone project is available at:

<https://github.com/KlariVZ/air-quality-risk-prediction>

The repository includes the raw data files, processed datasets, notebooks, figures, tables, and supporting project documentation used for the Interim Report.

## Air Quality Risk Prediction Using Environmental and Meteorological Data

---

### Introduction

#### Background and Context

Poor air quality can negatively affect environmental conditions, public health awareness, and local sustainability planning. It is therefore an important issue within the climate and sustainability domain. Prior research has linked air pollution, especially fine particulate matter such as PM<sub>2.5</sub>, to major health risks, which supports the continued importance of air quality monitoring and prediction (Wan Mahiyddin et al., 2023). Recent air quality research has also moved toward forward looking analytical tools rather than relying only on retrospective summaries of pollution conditions (Wang et al., 2025).

Air quality is influenced by multiple interacting factors, including pollutant concentrations, temperature, precipitation, wind speed, and seasonal conditions. These factors vary across time and location, which makes poor air quality difficult to anticipate without a structural predictive framework. Because air quality patterns emerge from measurable environmental and meteorological variables, predictive analytics is appropriate for examining how these variables relate to poor air quality days. Prior research also supports the use of meteorological variables and seasonal patterns in air quality prediction studies (Eren et al., 2025).

This project is relevant because predictive classification of poor air quality days can support more informed environmental monitoring, earlier risk awareness, and better decision making for planners, sustainability teams, public information units, and environmental agencies. Instead of relying only on after the fact observation, a predictive framework can help identify conditions under which local air quality is more likely to worsen. The usefulness of machine learning approaches for air quality management and early warning contexts has also been emphasized in recent AQI prediction research (Tırink, 2025).

### **Problem Statement**

Predicting poor air quality consistently remains difficult because poor air quality is shaped by multiple interacting pollutant and meteorological factors that change over time. Without a clear predictive framework, it is challenging for decision makers to identify the main drivers of poor air quality, understand seasonal variation, and select suitable models for the binary classification of poor air quality days. Prior literature also notes the need for models that balance predictive performance with practical utility and interpretability (Zhou et al., 2024).

### **Purpose of the Study**

The purpose of this study is to develop a predictive framework for the binary classification of poor air quality using environmental and meteorological data. Specifically, the study aims to identify key variables associated with poor air quality days, examine whether those days can be classified using pollutant and weather data, assess whether poor air quality days vary across seasons and time periods, and compare predictive model performance. This purpose aligns

with current research trends that combine exploratory analysis, temporal pattern review, and comparative model evaluation in air quality forecasting studies (Wang et al., 2025; Tırınk, 2025).

### **Interim Project Status (Progress Snapshot)**

- Completed: The raw datasets have been collected, cleaned, standardized, and merged into a daily analytical dataset.
- Completed: The study specific threshold for `poor\_air\_quality` was finalized as ` $\text{overall\_aqi} > 35$ `, based on threshold sensitivity review and class balance considerations.
- Completed: Exploratory data analysis was conducted, including descriptive statistics, target distribution review, pollutant specific AQI trends, seasonal and monthly summaries, correlation analysis, and dominant pollutant analysis.
- Completed: Preliminary classification models were trained and evaluated, including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine.
- In progress: Interpretation of model outputs, feature importance, cross-validation results, and final alignment of findings to the research questions.
- Pending: Formal inferential testing for seasonal and temporal variation, final model refinement, final recommendations, and completion of the final capstone report.

## **Scope and objectives**

The scope of this study is limited to the binary classification of poor air quality days and using publicly available environmental and meteorological data. The study focuses on Kapolei, Honolulu County, Hawaii, USA, for the period of January 1, 2024 to December 31, 2025. The unit of analysis is the daily observation, meaning each row in the final dataset represents one calendar date for the selected study setting. This scope keeps the project focused and feasible while still allowing for descriptive analysis, inferential testing, and predictive modeling.

The project uses U.S. EPA AirData / Air Quality System (AQS) daily air quality data and NOAA Climate Data Online (CDO) Daily Summaries for weather related variables. These data sources were merged by date to create one daily analytical dataset. The main target variable is `poor\_air\_quality`, which is derived from `overall\_aqi`. The study is therefore focused on the binary classification of poor air quality rather than multi-category AQI prediction.

The objectives of this study are to identify pollutant and meteorological variables associated with poor air quality days, evaluate whether poor air quality days can be classified using pollutant and weather data, examine whether poor air quality days vary across seasons and time periods, and compare candidate classification models to determine which model performs the best.

### **Research Question 1 (RQ1)**

Which environmental and meteorological variables are most strongly associated with poor air quality days?

### **Research Question 2 (RQ2)**

Can poor air quality days be accurately identified and classified using pollutant and weather data?

### **Research Question 3 (RQ3)**

Do poor air quality days vary significantly across seasons and time periods?

### **Research Question 4 (RQ4)**

Which predictive model performs best for the binary classification of poor air quality?

### **Literature survey**

Paraphrase the work done and state how each article is pertinent to your research. Present at least 10 relevant sources.

### **Literature Review Approach**

The literature reviewed for this interim report was selected to support the study's focus on air quality prediction, pollutant and meteorological predictors, seasonal variation, and machine learning model comparison. Sources were selected based on their relevance to at least one of the four research questions. Priority was given to peer reviewed articles that addressed air pollution risk, air quality index prediction, meteorological influences, seasonal air quality patterns, machine learning models, and model evaluation. The reviewed literature supports the project's use of pollutant data, weather variables, time based features, and comparative classification models.

## Summary of Key Literature

Recent literature shows that air quality prediction is commonly approached as a data driven problem that combines pollutant measurements, meteorological variables, temporal features, and machine learning models. This literature is relevant for this capstone because the study uses pollutant and weather data to classify poor air quality days and compare predictive models. The reviewed sources support the need for air quality prediction, the inclusion of meteorological predictors, the use of seasonal analysis, and the comparison of multiple modeling approaches.

Wan Mahiyuddin et al. (2023) reviewed the cardiovascular and respiratory health effects of fine particulate matter, especially PM<sub>2.5</sub>. This source is relevant because it supports the importance of air quality monitoring and predictions from a public health and environmental perspective. Although this capstone does not directly model health outcomes, the article helps explain why identifying poor air quality days is important.

Islam et al. (2025) also examined health impacts linked to air pollution, including cardiovascular effects. This source is useful for the background of the study because it reinforces the broader significance of air pollution as more than a technical measurement problem. It supports the argument that improved air quality awareness can be valuable for communities and decision makers.

Eren et al. (2025) studied seasonal patterns and machine learning based predictions of air pollutants in relation to meteorological parameters. This article is directly relevant because it connects air pollutants, weather variables, seasonality, and machine learning in one study. It

supports the use of meteorological predictors and helps justify RQ3, which examines whether poor air quality days vary across seasons and time periods.

Kamińska (2018) used random forests to model short term air pollution effects with traffic and meteorological conditions. This source is relevant because it supports the use of random forest methods for air quality modeling and shows that meteorological predictors can improve understanding of short term pollution patterns. It also supports the inclusion of feature importance interpretation in the current study.

Qi et al. (2024) compared multiple linear regression, generalized additive models, and random forest methods for meteorological normalization of PM<sub>2.5</sub>. This source is important because it demonstrates the value of comparing statistical and machine learning methods rather than relying on one model. It supports the capstone's plan to compare multiple candidate models for poor air quality classification.

Sachdeva et al. (2024) developed an AQI predictor using meteorological and pollutant concentration variables. This article is closely aligned with the current study because it uses both pollutant and meteorological inputs for AQI related prediction. It supports the decision to merge EPA pollutant data with NOAA weather data in one analytical dataset.

Sigamani and Venkatesan (2022) examined AQI prediction using meteorological parameters and machine learning for Internet of Things (IoT) applications. This source supports the practical application side of the capstone because it shows how air quality prediction can support monitoring systems and decision support tools. It is relevant to the project's recommendation and application focus.

Tirink (2025) compared machine learning methods, including support vector machine, for AQI forecasting under long term environmental patterns. This article is useful because it supports the idea that multiple models should be compared under a shared evaluation framework. It also helps justify the inclusion of support vector machine as one of the candidate classifiers.

Wang et al. (2025) discussed time series data driven PM2.5 forecasting and connected theoretical forecasting frameworks with empirical analysis. This source is relevant because the present project uses daily observations and time based features such as month, season, day of week, and weekend status. Although this capstone is focused on binary classification rather than full time series forecasting, the article supports the use of temporal structure in air quality modeling.

Zhou et al. (2024) reviewed deep learning architecture for PM2.5 prediction and emphasized the complexity of air pollution prediction. This source is useful because it highlights challenges in air quality modeling of air pollution prediction. This source is useful because it highlights challenges in air quality modeling, including model complexity, evaluation, and interpretability. For this capstone, the article supports the decision to prioritize feasible and interpretable methods rather than deep learning models that may be difficult to justify within the project's timeline.

**Table 1***Literature relevance matrix*

| <b>Author (Year)</b>         | <b>Domain/Context</b>                        | <b>Method(s)</b>                       | <b>Key Findings</b>                                                         |
|------------------------------|----------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------|
|                              |                                              |                                        | Supports the importance of monitoring and identifying poor air quality days |
| Wan Mahiyuddin et al. (2023) | PM2.5 and health affects                     | Literature review                      | Supports the public health relevance of air quality awareness               |
| Islam et al. (2025)          | Air pollution and cardiovascular disease     | Systematic review                      | Supports use of weather variables and seasonal analysis for RQ3             |
| Eren et al. (2025)           | Air pollutants, meteorology, and seasonality | Machine learning and seasonal analysis | Supports Random Forest and feature importance interpretation                |
| Kamińska (2018)              | Short term air pollution modeling            | Random forest                          | Supports comparing statistical and machine learning approaches              |
|                              | PM2.5 and meteorological normalization       | Regression, GAM, Random Forest         |                                                                             |
| Qi et al. (2024)             |                                              |                                        |                                                                             |

|                              |                                       |                                         |                                                   |
|------------------------------|---------------------------------------|-----------------------------------------|---------------------------------------------------|
|                              |                                       | Pollutant and meteorological predictors | Supports merging pollutant and weather variables  |
| Sachdeva et al. (2024)       | AQI prediction                        |                                         |                                                   |
| Sigamani & Venkatesan (2022) | AQI prediction for monitoring systems | Machine learning                        | Supports practical decision support               |
|                              |                                       |                                         | Supports comparative modeling and inclusion of    |
| Tirink (2025)                | AQI forecasting                       | XGBoost, LightGBM, SVM                  | SVM                                               |
|                              |                                       |                                         | Supports daily observations and temporal features |
| Wang et al. (2025)           | PM2.5 forecasting                     | Time series data driven methods         | Supports concern for                              |
|                              | PM2.5 prediction                      | Deep learning                           | evaluation, complexity,                           |
| Zhou et al. (2024)           | complexity                            | review                                  | and interpretability                              |

Overall, the literature supports the design of this study in four ways. First, it shows that poor air quality is an important environmental and public health concern. This supports the practical importance of identifying poor air quality days, even though the present study does not directly model health outcomes.

Second, the literature supports the inclusion of both pollutant and meteorological variables. Several reviewed studies use pollutant concentrations together with weather conditions to improve air quality prediction. This supports the current study's decision to merge Environmental Protection Agency (EPA) pollutant data and National Oceanic and Atmospheric Administration (NOAA) meteorological data.

Third, the literature supports examining seasonal and time based variation. Prior work has shown that air pollution patterns may vary by season and environmental conditions. This supports the inclusion of time features such as month, season, day of week, and weekend status.

Fourth, the literature supports comparing multiple predictive models rather than relying on one method. The reviewed studies show that air quality modeling can involve regression based methods, tree based methods, support vectors machines, and more advanced machine learning approaches. For this study, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine provide a feasible and interpretable model comparison framework.

## **Data Description**

### **Data Source(s) and Access**

This study uses publicly available air quality and meteorological data to classify poor air quality days for Kapolei, Honolulu County, Hawaii. The air quality data were obtained from U.S. EPA AirData / Air Quality System (AQS) daily files, and the weather data were obtained from NOAA Climate Data Online (CDO) Daily Summaries. These sources are appropriate because the

project requires daily pollutant measures, pollutant specific Air Quality Index (AQI) values, and daily weather variables for the same study period.

The raw EPA files include separate daily pollutant files for nitrogen dioxide (NO<sub>2</sub>), ozone, PM<sub>10</sub>, and PM<sub>2.5</sub>. The EPA files include daily meteorological observations from Honolulu International Airport, HI US, which was selected as the available nearby weather station with complete meteorological variables for the study period. The raw and processed data files, code notebooks, figures, and output tables will be available in the GitHub repository for access.

## **Dataset Overview**

The final cleaned dataset covers the period January 1, 2024 through December 31, 2025. The unit of analysis is the daily observation, meaning each row represents one calendar date. After cleaning and merging the EPA and NOAA files by date, the final cleaned dataset contains 731 daily observations and 41 variables.

The main target variable is `poor\_air\_quality`. This variable is derived from `overall\_aqi`, which is calculated as the maximum available pollutant specific AQI value for each date. For this interim report, `poor\_air\_quality` is operationalized as a binary variable where days with `overall\_aqi > 35` are classified as poor air quality days. This threshold is a study specific definition of locally elevated AQI conditions and is not presented as an official EPA health category cutoff.

The final target distribution includes 581 non-poor air quality days and 150 poor air quality days. This means that poor air quality days represent 20.52% of the final cleaned dataset.

**Number of records (rows):** 731 daily observations

**Number of variables (columns):** 41 variables

**Time period:** January 1, 2024 to December 31, 2025

**Unit of analysis:** Daily observation

**Study setting:** Kapolei, Honolulu County, Hawaii, USA

**Air quality source:** EPA Airdata / AQS daily data

**Weather source:** NOAA CDO Daily Summaries

**Target variable(s):** `poor\_air\_quality`

**Target definition:** `poor\_air\_quality = 1` when `overall\_aqi > 35`; otherwise `0`

### **Data Dictionary (Mandatory)**

The main variables used in the cleaned dataset are summarized in Table 2. The full cleaned dataset also retains additional site metadata, pollutant specific AQI values, observation count fields, percent complete fields, and parameter occurrence code fields for documentation and quality reviews.

**Table 2***Data dictionary*

| <b>Variable</b> | <b>Definition</b>                                                         | <b>Data</b>      | <b>Allowed Values</b>    | <b>Missing</b>  | <b>Notes</b>           |
|-----------------|---------------------------------------------------------------------------|------------------|--------------------------|-----------------|------------------------|
| <b>Name</b>     |                                                                           | <b>Type</b>      | <b>/ Range</b>           | <b>Values</b>   |                        |
|                 |                                                                           |                  |                          | <b>Handling</b> |                        |
| date            | Calendar date of observation                                              | Date             | 2024-01-01 to 2025-12-31 | YYYY-M<br>M-DD  | Merge key / Identifier |
| site_id         | Unique EPA monitoring site identifier for the Kapolei monitoring location | String / Integer | 150030010                | Not applicable  | Identifier             |
| local_site_name | Name of the air monitoring site                                           | String           | Kapolei                  | Not applicable  | Identifier             |
| county          | County in which the monitoring site is located                            | String           | Honolulu                 | Not applicable  | Identifier             |

|                     |                                                     |        |                    |                 |               |
|---------------------|-----------------------------------------------------|--------|--------------------|-----------------|---------------|
| state               | State in which the monitoring site is located       | String | Hawaii             | Not applicable  | Identifier    |
| site_latitude       | Latitude of the EPA monitoring site                 | Float  | Decimal degrees    | Decimal degrees | Site metadata |
| site_longitude      | Longitude of the EPA monitoring site                | Float  | Decimal degrees    | Decimal degrees | Site metadata |
| daily_max_1hr_no2   | Daily maximum 1-hour nitrogen dioxide concentration | Float  | EPA                | ppb             | Predictor     |
| daily_max_8hr_ozone | Daily maximum 8-hour ozone concentration            | Float  | Observed EPA range | ppm             | Predictor     |
| daily_mean_pm25     | Daily mean PM2.5 concentration                      | Float  | Observed EPA range | ug/m3           | Predictor     |
| daily_mean_pm10     | Daily mean PM10 concentration                       | Float  | Observed EPA range | ug/m3           | Predictor     |

|                           |                                                                        |         |                       |              |                          |
|---------------------------|------------------------------------------------------------------------|---------|-----------------------|--------------|--------------------------|
| daily_aqi_no<br>2         | Daily AQI value<br>associated with NO2                                 | Integer | Observed EPA<br>range | AQI<br>index | Intermediate<br>variable |
| daily_aqi_oz<br>one       | Daily AQI value<br>associated with ozone                               | Integer | Observed EPA<br>range | AQI<br>index | Intermediate<br>variable |
| daily_aqi_pm<br>25        | Daily AQI value<br>associated with PM2.5                               | Integer | Observed EPA<br>range | AQI<br>index | Intermediate<br>variable |
| daily_aqi_pm<br>10        | Daily AQI value<br>associated with PM10                                | Integer | Observed EPA<br>range | AQI<br>index | Intermediate<br>variable |
| daily_obs_co<br>unt_no2   | Number of<br>observations used to<br>derive daily NO2<br>summary value | Integer | Observed EPA<br>range | Count        | Data quality<br>variable |
| daily_obs_co<br>unt_ozone | Number of<br>observations used to<br>derive daily ozone<br>summary     | Integer | Observed EPA<br>range | Count        | Data quality<br>variable |
| daily_obs_co              | Number of                                                              | Integer | Observed EPA          | Count        | Data quality             |

|              |                                                             |         |              |         |              |
|--------------|-------------------------------------------------------------|---------|--------------|---------|--------------|
| unt_pm25     | observations used to<br>derive daily PM2.5<br>summary value |         | range        |         | variable     |
| daily_obs_co | Number of                                                   | Integer | Observed EPA | Count   | Data quality |
| unt_pm10     | observations used to<br>derive daily PM10<br>summary value  |         | range        |         | variable     |
| percent_com  | Percentage                                                  | Float   | Observed EPA | Percent | Data quality |
| plete_no2    | completeness of NO2<br>observations for the<br>day          |         | range        |         | variable     |
| percent_com  | Percentage                                                  | Float   | Observed EPA | Percent | Data quality |
| plete_ozone  | completeness of ozone<br>observations for the<br>day        |         | range        |         | variable     |
| percent_com  | Percentage                                                  | Float   | Observed EPA | Percent | Data quality |
| plete_pm25   | completeness of<br>PM2.5 observations                       |         | range        |         | variable     |

|              |                      |        |                |           |              |
|--------------|----------------------|--------|----------------|-----------|--------------|
|              | for the day          |        |                |           |              |
| percent_com  | Percentage           | Float  | Observed EPA   | Percent   | Data quality |
| plete_pm10   | completeness of      |        | range          |           | variable     |
|              | PM10 observations    |        |                |           |              |
|              | for the day          |        |                |           |              |
| station      | NOAA station         | String | USW00022521    | Not       | Identifier   |
|              | identifier for the   |        |                | applicabl |              |
|              | selected weather     |        |                | e         |              |
|              | station              |        |                |           |              |
| station_name | Name of the selected | String | Honolulu       | Not       | Site         |
|              | NOAA weather         |        | International  | applicabl | metadata     |
|              | station              |        | Airport, HI US | e         |              |
| latitude     | Latitude of the NOAA | Float  | NOAA           | Decimal   | Site         |
|              | weather station      |        |                | degrees   | metadata     |
| longitude    | Longitude of the     | Float  | NOAA           | Decimal   | Site         |
|              | NOAA weather         |        |                | degrees   | metadata     |
|              | station              |        |                |           |              |

|           |                                       |         |                             |                   |               |
|-----------|---------------------------------------|---------|-----------------------------|-------------------|---------------|
| elevation | Elevation of the NOAA weather station | Float   | NOAA                        | Meters            | Site metadata |
| tmax      | Daily maximum temperature             | Float   | Degrees Celsius             | Degrees Celsius   | Predictor     |
| tmin      | Daily minimum temperature             | Float   | Degrees Celsius             | Degrees Celsius   | Predictor     |
| prcp      | Daily precipitation                   | Float   | Metric units from NOAA file | Millimetres       | Predictor     |
| awnd      | Average daily wind speed              | Float   | Metric units from NOAA file | Meters per second | Predictor     |
| year      | Year extracted from date              | Integer | 2024 or 2025                | Year              | Time feature  |
| month     | Month extracted from date             | Integer | 1 to 12                     | 1 to 12           | Time feature  |
| season    | Season category                       | String  | Winter, Spring,             | Not               | Time feature  |

|                      |                                                                                                    |                     |                                |                              |                         |
|----------------------|----------------------------------------------------------------------------------------------------|---------------------|--------------------------------|------------------------------|-------------------------|
|                      | assigned to the<br>observation date                                                                |                     | Summer, Fall                   | applicabl<br>e               |                         |
| day_of_week          | Day of the week<br>derived from date                                                               | String /<br>Integer | Monday<br>through Sunday       | Not<br>applicabl<br>e        | Time feature            |
| is_weekend           | Indicator showing<br>whether the<br>observation date falls<br>on a weekend                         | Integer             | 0 = weekday, 1<br>= weekend    | 0 = No, 1<br>= Yes           | Time feature            |
| overall_aqi          | Daily overall AQI<br>calculated as the<br>maximum AQI across<br>pollutant specific AQI<br>values   | Integer             | 12 to 56 in<br>cleaned dataset | AQI<br>index                 | Outcome<br>construction |
| poor_air_qual<br>ity | Binary target variable<br>indicating whether the<br>day is classified as a<br>poor air quality day | Binary              | 0 = Not poor, 1<br>= Poor      | 0 = Not<br>poor, 1 =<br>Poor | Target<br>variable      |

|                      |                                                          |        |                         |                               |                             |
|----------------------|----------------------------------------------------------|--------|-------------------------|-------------------------------|-----------------------------|
| `dominant_pollutant` | Pollutant with the highest AQI contribution for the date | String | Ozone, PM2.5, PM10, NO2 | Based on available AQI values | Used for EDA interpretation |
|----------------------|----------------------------------------------------------|--------|-------------------------|-------------------------------|-----------------------------|

### GitHub Data Availability Statement

The project files are organized in the GitHub repository titled `air-quality-risk-prediction`. The repository will provide access to the raw data, cleaned data, code notebooks, figures, tables, and supporting documentation. A screenshot, Figure 1, shows a preview of the cleaned dataset.

**Raw data:** `data/raw/`

**Processed/clean data:** `data/processed/`

**Code/notebooks:** `notebooks/`

**Screenshots / Evidence (as required)**

**Figure 1***Dataset Preview Screenshot*

| date       | site_id   | local_site_name | county   | state  | site_latitude | site_longitude | station      | station_name                          | daily_max_6hr_ozone | daily_aqi_ozone | daily_obs_count_ozone | percent_complete_ozone | ozone_poc | daily_mean_pm10 | d |
|------------|-----------|-----------------|----------|--------|---------------|----------------|--------------|---------------------------------------|---------------------|-----------------|-----------------------|------------------------|-----------|-----------------|---|
| 2024-01-01 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.033               | 31.0            | 17.0                  | 100.0                  | 1         | 13.0            | 1 |
| 2024-01-02 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.035               | 32.0            | 17.0                  | 100.0                  | 1         | 20.0            | 1 |
| 2024-01-03 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.035               | 32.0            | 17.0                  | 100.0                  | 1         | 25.0            | 2 |
| 2024-01-04 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.033               | 31.0            | 17.0                  | 100.0                  | 1         | 21.0            | 1 |
| 2024-01-05 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.031               | 29.0            | 17.0                  | 100.0                  | 1         | 23.0            | 2 |
| 2024-01-06 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.034               | 31.0            | 17.0                  | 100.0                  | 1         | 17.0            | 1 |
| 2024-01-07 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.032               | 30.0            | 17.0                  | 100.0                  | 1         | 15.0            | 1 |
| 2024-01-08 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.03                | 28.0            | 17.0                  | 100.0                  | 1         | 22.0            | 2 |
| 2024-01-09 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.037               | 34.0            | 17.0                  | 100.0                  | 1         | 18.0            | 1 |
| 2024-01-10 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.039               | 36.0            | 17.0                  | 100.0                  | 1         | 14.0            | 1 |
| 2024-01-11 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US |                     |                 |                       |                        |           | 17.0            | 1 |
| 2024-01-12 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.037               | 34.0            | 16.0                  | 94.0                   | 1         | 15.0            | 1 |
| 2024-01-13 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.043               | 40.0            | 17.0                  | 100.0                  | 1         | 24.0            | 2 |
| 2024-01-14 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.041               | 38.0            | 17.0                  | 100.0                  | 1         | 18.0            | 1 |
| 2024-01-15 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.036               | 33.0            | 17.0                  | 100.0                  | 1         | 12.0            | 1 |
| 2024-01-16 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.035               | 32.0            | 17.0                  | 100.0                  | 1         | 22.0            | 2 |
| 2024-01-17 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.038               | 35.0            | 13.0                  | 76.0                   | 1         | 36.0            | 3 |
| 2024-01-18 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.029               | 27.0            | 17.0                  | 100.0                  | 1         | 23.0            | 2 |
| 2024-01-19 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.017               | 16.0            | 17.0                  | 100.0                  | 1         | 11.0            | 1 |
| 2024-01-20 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.015               | 14.0            | 17.0                  | 100.0                  | 1         | 12.0            | 1 |
| 2024-01-21 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.019               | 18.0            | 17.0                  | 100.0                  | 1         | 16.0            | 1 |
| 2024-01-22 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.023               | 21.0            | 17.0                  | 100.0                  | 1         | 21.0            | 1 |
| 2024-01-23 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.023               | 21.0            | 17.0                  | 100.0                  | 1         | 18.0            | 1 |
| 2024-01-24 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.025               | 23.0            | 17.0                  | 100.0                  | 1         | 22.0            | 2 |
| 2024-01-25 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.032               | 30.0            | 17.0                  | 100.0                  | 1         | 18.0            | 1 |
| 2024-01-26 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.024               | 22.0            | 17.0                  | 100.0                  | 1         | 14.0            | 1 |
| 2024-01-27 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.016               | 15.0            | 17.0                  | 100.0                  | 1         | 13.0            | 1 |
| 2024-01-28 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.018               | 17.0            | 17.0                  | 100.0                  | 1         | 20.0            | 1 |
| 2024-01-29 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.019               | 18.0            | 17.0                  | 100.0                  | 1         | 31.0            | 2 |
| 2024-01-30 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.03                | 28.0            | 17.0                  | 100.0                  | 1         | 19.0            | 1 |
| 2024-01-31 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.034               | 31.0            | 17.0                  | 100.0                  | 1         | 18.0            | 1 |
| 2024-02-01 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.03                | 28.0            | 17.0                  | 100.0                  | 1         | 14.0            | 1 |
| 2024-02-02 | 150030010 | Kapolei         | Honolulu | Hawaii | 21.32374479   | -158.0886127   | USW000022521 | HONOLULU INTERNATIONAL AIRPORT, HI US | 0.045               | 42.0            | 17.0                  | 100.0                  | 1         | 17.0            | 1 |

**Analysis**

The analysis stage focused on preparing the raw EPA and NOAA datasets for daily level exploratory analysis and preliminary modeling. This section describes the cleaning process, minimum sample size justification, exploratory data analysis (EDA), and the main insights identified from the cleaned dataset.

**Data Cleaning**

The raw data files consisted of four EPA pollutant datasets and one NOAA meteorological dataset. The EPA files included daily records for nitrogen dioxide (NO<sub>2</sub>), ozone, PM<sub>10</sub>, and PM<sub>2.5</sub> from the Kapolei monitoring site. The NOAA file included daily

meteorological records from Honolulu International Airport, HI US, which was used as the available nearby weather station for the study period. The final analysis period was January 1, 2024, through December 31, 2025, with each row representing one calendar date.

The first cleaning step was to standardize all date fields. The EPA files used a month/day/year format, while the NOAA file used a year-month-day format. These were converted into a common `date` variable so that all files could be merged by daily observation. Next, relevant variables were selected from each file, including pollutant concentration values, pollutant specific Air Quality Index (AQI) values, observation counts, percent complete fields, latitude, and longitude, were retained for documentation.

The PM2.5 file required additional cleaning because it contained multiple records for some dates due to different parameter occurrence code (POC) values. To preserve the daily unit of analysis, duplicate PM2.5 dates were aggregated into one record per date. Daily PM2.5 concentration was averaged, while the daily PM2.5 AQI value was summarized using the maximum value for that date. This approach preserved the highest AQI burden for daily risk classification while keeping the dataset aligned to one row per date.

After cleaning each file, a complete daily calendar was created for the full study period. The cleaned EPA and NOAA files were then merged into this calendar by `date`. This produced a final cleaned dataset with 731 daily observations and 41 variables, covering the full period from 2024-01-01 to 2025-12-31. The merged dataset contained zero duplicate dates, confirming that the unit analysis was correctly structured as one row per day.

Missing values were reviewed after merging. The largest missingness occurred in the pollutant files, especially NO<sub>2</sub>, PM<sub>10</sub>, and PM<sub>2.5</sub>, because not every pollutant file contained a record for every date. NOAA weather variables had only two missing dates. For EDA, missing values were later handled using median imputation in the modeling notebook.

The derived variable `overall\_aqi` was calculated as the maximum available pollutant specific AQI value for each date. The binary target variable `poor\_air\_quality` was then created from `overall\_aqi`. A threshold sensitivity review was conducted before finalizing the target definition. The original draft threshold of `overall\_aqi > 50` identified only 12 poor air quality days, or 1.64% of the dataset. Because this would create severe class imbalance, the study specific threshold was set to `overall\_aqi > 35`, which identified 150 poor air quality days, or 20.52% of the dataset. This threshold is used as an operational definition of locally elevated AQI conditions within the Kapolei study period, not as an official EPA health category cutoff.

**Table 3**

*Data cleaning log*

| Variables                 |                            | Detection              |                          |                                    |
|---------------------------|----------------------------|------------------------|--------------------------|------------------------------------|
| Issue                     | Affected                   | Method                 | Treatment Applied        | Rationale                          |
|                           |                            | Column                 | Converted all dates      | Required for                       |
| Inconsistent date formats | EPA `Date`;<br>NOAA `DATE` | inspection and parsing | to a common `date` field | merging files by daily observation |

|                   |                   |                |                    |                   |
|-------------------|-------------------|----------------|--------------------|-------------------|
|                   |                   |                |                    | Preserved         |
|                   |                   |                |                    | pollutant         |
|                   |                   |                | Cleaned each       | specific fields   |
|                   |                   |                | pollutant file     | and avoided       |
| Multiple EPA      | NO2, ozone,       |                | separately before  | column            |
| pollutant files   | PM10, PM2.5       | File inventory | merging            | confusion         |
|                   |                   |                |                    | Maintained one    |
|                   |                   |                |                    | row per date      |
| Duplicate PM2.5   | PM2.5             |                |                    | while preserving  |
| dates due to POC  | concentration and | Duplicate date | Aggregated to one  | daily AQI         |
| values            | AQI               | review         | daily PM2.5 record | burden            |
|                   |                   |                | Retained during    | Reflected true    |
|                   |                   |                | EDA; median        | data availability |
|                   |                   |                | imputation planned | while enabling    |
| Missing pollutant | NO2, PM10,        | Missing value  | for modeling       | preliminary       |
| dates             | PM2.5, ozone      | summary        | predictors         | modeling          |
|                   |                   |                | EDA; median        |                   |
|                   |                   |                | imputation planned | Only tow          |
| Missing weather   | `awnd`, `prcp`,   | Missing value  | for modeling       | weather dates     |
| dates             | `tmax`, `tmin`    | summary        | predictors         | were missing      |

|                  |                    |                    |                       |                   |
|------------------|--------------------|--------------------|-----------------------|-------------------|
|                  |                    |                    |                       | Produced a        |
|                  |                    |                    |                       | feasible positive |
|                  |                    |                    |                       | class while       |
|                  |                    |                    |                       | representing      |
| Target threshold | `poor_air_quality` | Threshold          | Finalized threshold   | locally elevated  |
| uncertainty      | `                  | sensitivity review | as `overall_aqi > 35` | AQI days          |
|                  |                    |                    |                       | Ensured each      |
|                  |                    |                    | Confirmed zero        | row represented   |
| Daily unit of    | `date`             | Duplicate date     | duplicate dates after | one calendar      |
| analysis         |                    | check              | merge                 | date              |

### Minimum Sample Size Computation per Research Question

Minimum sample size was reviewed for each research question using either a power-based or confidence-interval based approach. The largest required minimum sample size was selected as the overall minimum analytical requirement. The final cleaned dataset contains 731 daily observations, which exceeds the required minimum sample size of 385 observations and support analysis, train/test partitioning, and preliminary modeling comparison.

**Table 4***Minimum Sample Size Computation by Research Question*

| Research |                                             |                                                 | Minimum Sample |
|----------|---------------------------------------------|-------------------------------------------------|----------------|
| Question | Method Used                                 | Key Parameters                                  | Size (N)       |
| RQ1      | Power/Green's Rule for multiple regression. | $\alpha = 0.05$ , predictors = 8                | 114            |
|          |                                             | $Z = 1.96$ , $e = 0.05$ , $p = 0.50$            |                |
| RQ2      | Confidence interval for proportion.         | $\alpha = 0.05$ , power = 0.80, effect size $f$ | 385            |
|          |                                             |                                                 |                |
| RQ3      | Power for one way ANOVA.                    | $\alpha = 0.05$ , groups = 4                    | 180            |
|          | Confidence interval for                     | $Z = 1.96$ , $e = 0.05$ , expected              |                |
| RQ4      | expected model accuracy.                    | accuracy = 0.80                                 | 246            |

The final minimum sample size requirement was therefore:  $\max(114, 385, 180, 246) = 385$ .

Because the cleaned dataset contains 731 daily observations, the available sample is adequate for the planned interim analysis. The dataset exceeds the minimum requirement while retaining the full daily study period.

## EDA Results (Interpretation Emphasis)

Exploratory data analysis was conducted to understand the distribution of AQI values, pollutant patterns, meteorological variables, seasonal variation, and relationships among key predictors. The main EDA outputs included descriptive statistics, target distribution review, dominant pollutant analysis, time series plots, seasonal summaries, monthly summaries, boxplots, and a correlation heatmap.

### Descriptive Summary of Key Variables

**Table 5**

*Descriptive Statistics for Key Variables*

| Variable     | Standard |       |           |         |        |         |
|--------------|----------|-------|-----------|---------|--------|---------|
|              | Count    | Mean  | Deviation | Minimum | Median | Maximum |
| Overall AQI  | 731      | 29.40 | 7.83      | 12.00   | 29.00  | 56.00   |
| Daily 8 hour |          |       |           |         |        |         |
| ozone        | 726      | 0.03  | 0.01      | 0.01    | 0.03   | 0.05    |
| Daily mean   |          |       |           |         |        |         |
| PM10         | 686      | 14.00 | 4.29      | 4.29    | 13.00  | 36.00   |
| Daily mean   |          |       |           |         |        |         |
| PM2.5        | 699      | 4.19  | 1.50      | 1.50    | 3.80   | 12.00   |
| Daily max 1  | 682      | 6.99  | 4.26      | 4.26    | 5.80   | 30.80   |

hour NO2

Average wind

|       |     |      |      |      |      |      |
|-------|-----|------|------|------|------|------|
| speed | 729 | 4.74 | 1.59 | 1.59 | 4.80 | 8.80 |
|-------|-----|------|------|------|------|------|

|               |     |      |      |      |      |       |
|---------------|-----|------|------|------|------|-------|
| Precipitation | 729 | 0.90 | 5.67 | 5.67 | 0.00 | 90.70 |
|---------------|-----|------|------|------|------|-------|

Maximum

|             |     |       |      |      |       |       |
|-------------|-----|-------|------|------|-------|-------|
| temperature | 729 | 29.53 | 1.86 | 1.86 | 30.00 | 33.30 |
|-------------|-----|-------|------|------|-------|-------|

Minimum

|             |     |       |      |      |       |       |
|-------------|-----|-------|------|------|-------|-------|
| temperature | 729 | 22.44 | 2.07 | 2.07 | 22.80 | 26.10 |
|-------------|-----|-------|------|------|-------|-------|

As shown in Table 5, overall AQI values were generally low to moderate within the local study period, ranging from 12 to 56. The mean and median overall AQI values were both close to 29, suggesting that most days were not high AQI days. This finding directly informed the target threshold decision because a conventional higher threshold would produce very few cases for classification.

## Target Distribution

**Figure 2**

*Distribution of Binary Poor Air Quality Target Using Study Threshold `overall\_aqi > 35`*

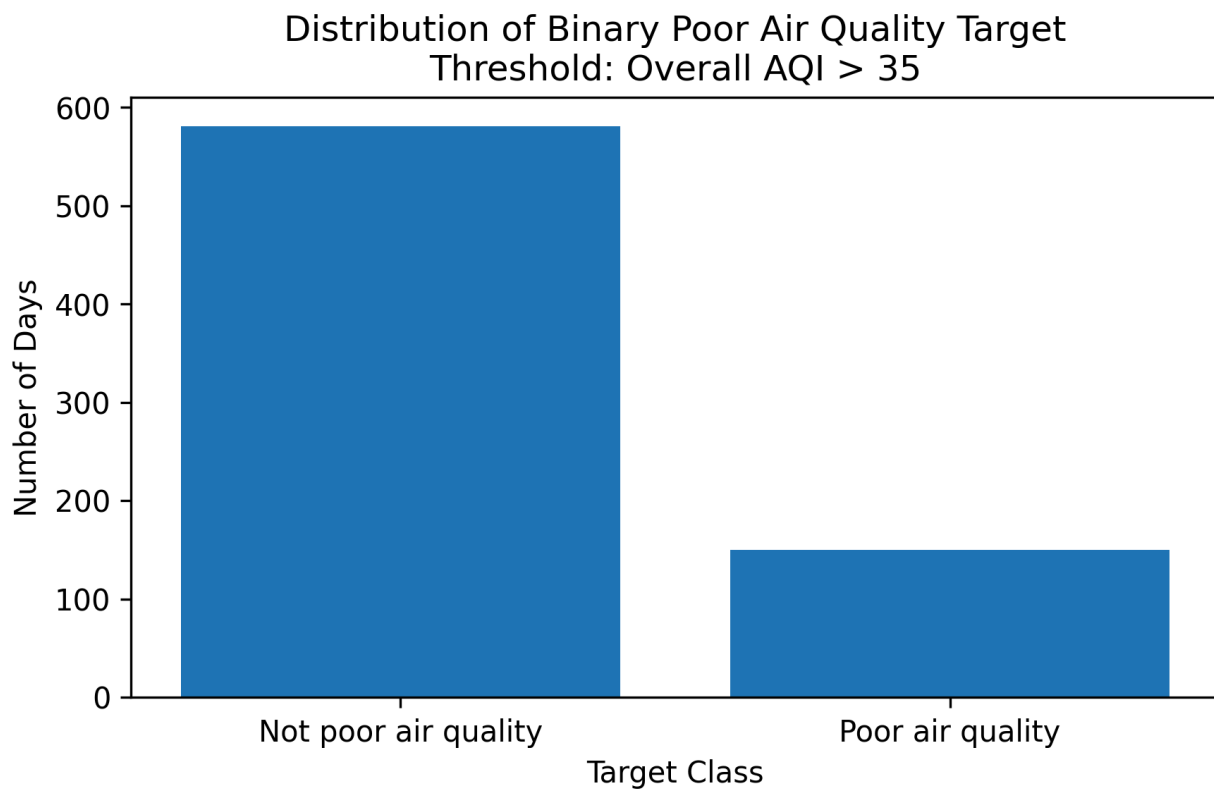


Figure 2 shows the final distribution of the binary target variable. Using the study specific threshold of `overall\_aqi > 35`, the dataset contained 581 non-poor air quality days and 150 poor air quality days. This means that poor air quality days represented 20.52% of the dataset. The distribution remains moderately imbalanced, but it is feasible for preliminary classification modeling. This threshold provides a more practical modeling target than the original draft threshold of `overall\_aqi > 50`, which identified only 12 poor air quality days.

**Table 6***Threshold Sensitivity Review for `poor\_air\_quality`*

| <b>Threshold Rule</b> | <b>Not poor</b> |                  |                      | <b>Selected</b>  |
|-----------------------|-----------------|------------------|----------------------|------------------|
|                       | <b>days</b>     | <b>Poor Days</b> | <b>Poor Days (%)</b> | <b>Threshold</b> |
| `overall_aqi > 30`    | 414             | 317              | 43.37                | No               |
| `overall_aqi > 35`    | 581             | 150              | 20.52                | Yes              |
| `overall_aqi > 40`    | 668             | 63               | 8.62                 | No               |
| `overall_aqi > 45`    | 707             | 24               | 3.28                 | No               |
| `overall_aqi > 50`    | 719             | 12               | 1.64                 | No               |

Table 6 shows that the number of poor air quality days changes substantially depending on the threshold. The selected threshold of `overall\_aqi > 35` was chosen because it identifies locally elevated AQI days while preserving enough positive cases for classification. This threshold supports the binary modeling objective while avoiding the severe imbalance created by higher thresholds.

## Overall AQI Over Time

**Figure 3**

*Overall AQI Over Time with Study Threshold*

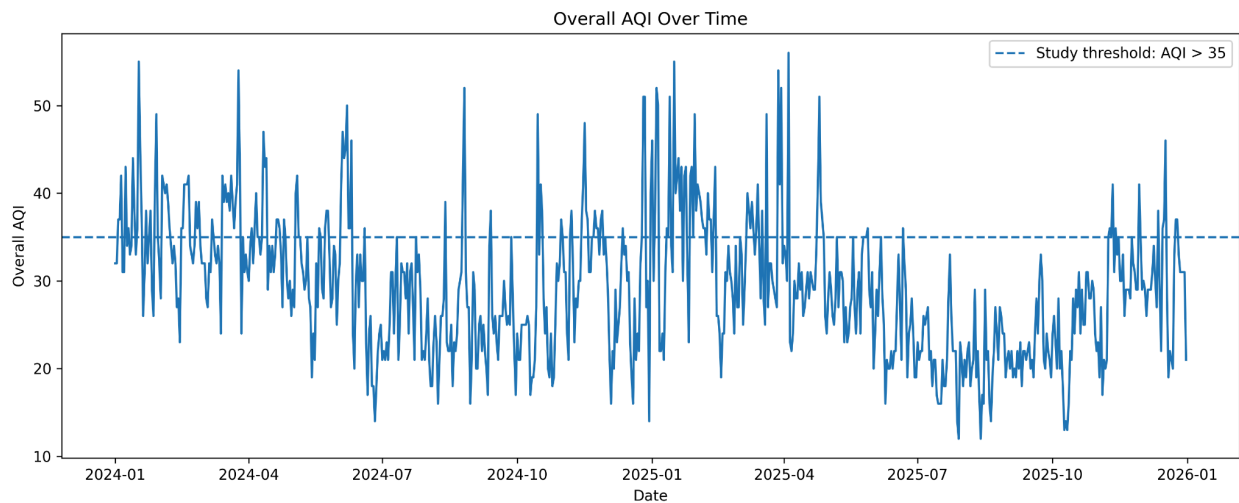


Figure 3 shows daily `overall\_aqi` values over the study period. The AQI values fluctuate over time, with several periods exceeding the study specific threshold of 35. These elevated periods appear more common in winter and spring months, which supports further seasonal analysis for RQ3. The time series pattern also supports the use of temporal features such as month, season, day of week, and weekend status.

## Pollutant Specific AQI Over Time

**Figure 4**

*Pollutant Specific AQI Values Over Time*

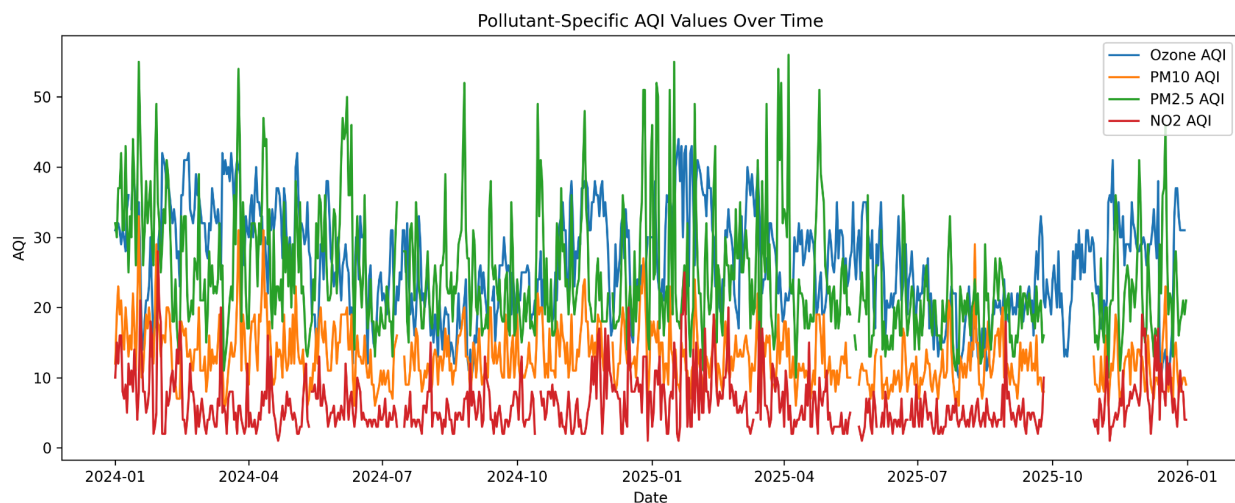


Figure 4 compares pollutant specific AQI values for ozone, PM10, PM2.5 and NO2. Ozone and PM2.5 appear to contribute most strongly to variation in daily AQI, while NO2 remains comparatively low across the study period. This visual pattern is consistent with the dominant pollutant analysis and the correlation results. The figure supports RQ1 by helping identify which pollutants appear most associated with elevated AQI conditions.

**Table 7***Dominant Pollutant Counts Based on Daily Overall AQI*

| <b>Dominant Pollutant</b> | <b>Number of Days</b> |
|---------------------------|-----------------------|
| Ozone                     | 474                   |
| PM2.5                     | 256                   |
| PM10                      | 1                     |
| NO2                       | 0                     |

Table 7 shows that ozone was the most frequent dominant pollutant, driving the highest pollutant specific AQI on 474 days. PM2.5 was the dominant pollutant on 256 days, while PM10 dominated on only one day and NO2 did not dominate on any day. This finding suggests that ozone and PM2.5 are especially important pollutants in the Kapolei dataset and should be emphasized in the interpretation and modeling.

## Seasonal Patterns

**Figure 5**

*Overall AQI by Season*

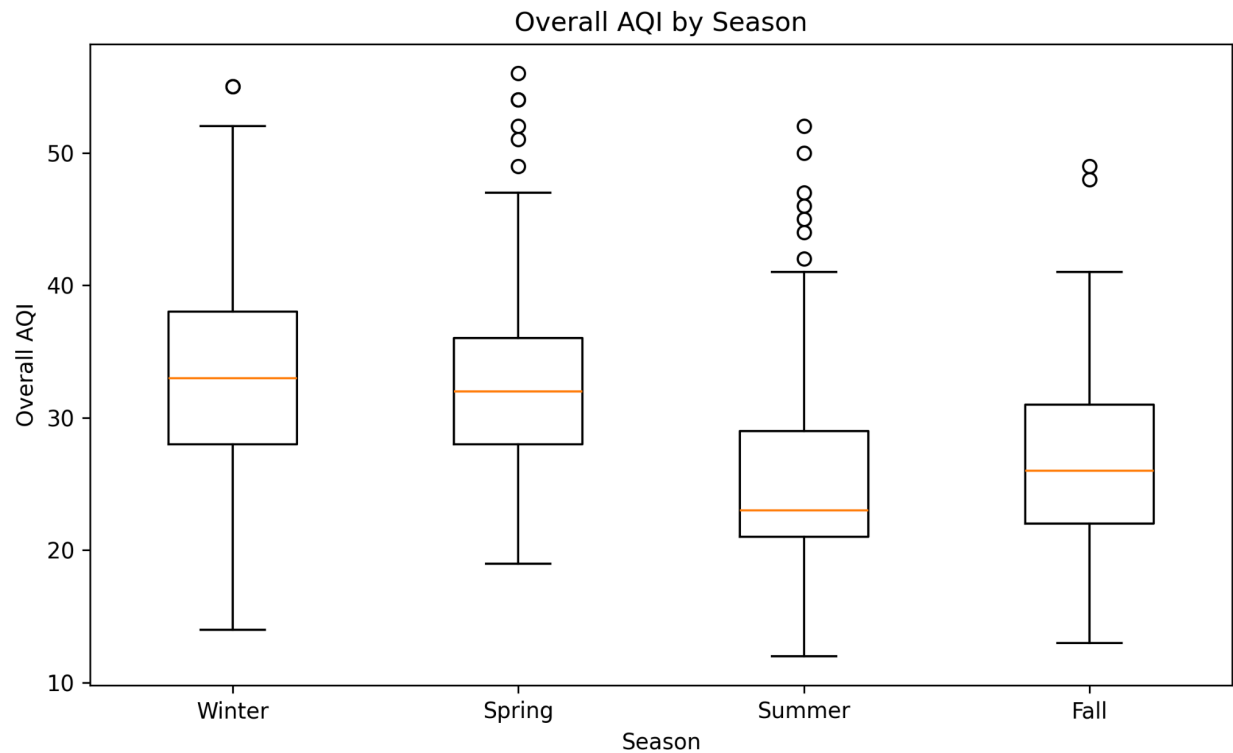


Figure 5 shows the distribution of overall AQI by season. The seasonal boxplot suggests that winter and spring generally had higher AQI distributions than summer and fall. This finding supports the relevance of RQ3, which asks whether poor air quality days vary across seasons and time periods.

**Table 8***Seasonal Summary of Overall AQI and Poor Air Quality Days*

| Season | Days | Mean    |                    | Maximum     | Poor Air     |
|--------|------|---------|--------------------|-------------|--------------|
|        |      | Overall | Median Overall AQI | Overall AQI | Quality Days |
| Fall   | 182  | 26.89   | 26.00              | 49.00       | 24           |
| Spring | 184  | 32.64   | 32.00              | 56.00       | 55           |
| Summer | 184  | 24.93   | 23.00              | 52.00       | 17           |
| Winter | 181  | 33.19   | 33.00              | 55.00       | 54           |

Table 8 shows that winter had the highest mean overall AQI at 33.19, followed closely by spring at 32.64. Spring had 55 poor air quality days, and winter had 54 poor air quality days. Summer had the lowest mean overall AQI at 24.93. These results suggest meaningful seasonal variation in air quality patterns and support further inferential testing in the final report.

## Monthly Patterns

**Table 9**

*Monthly Summary of Overall AQI and Poor Air Quality Days*

| Month | Days | Mean           |                    | Maximum<br>Overall AQI | Poor Air<br>Quality Days |
|-------|------|----------------|--------------------|------------------------|--------------------------|
|       |      | Overall<br>AQI | Median Overall AQI |                        |                          |
| 1     | 62   | 36.82          | 36.00              | 55.00                  | 35                       |
| 2     | 57   | 33.79          | 34.00              | 43.00                  | 16                       |
| 3     | 62   | 34.89          | 34.00              | 54.00                  | 26                       |
| 4     | 60   | 32.92          | 32.00              | 56.00                  | 20                       |
| 5     | 62   | 30.11          | 29.00              | 42.00                  | 9                        |
| 6     | 60   | 27.40          | 25.50              | 50.00                  | 5                        |
| 7     | 62   | 24.13          | 22.50              | 35.00                  | 0                        |
| 8     | 62   | 23.35          | 22.00              | 52.00                  | 1                        |
| 9     | 60   | 23.88          | 22.50              | 38.00                  | 3                        |
| 10    | 62   | 25.18          | 25.00              | 49.00                  | 4                        |
| 11    | 60   | 31.67          | 32.00              | 48.00                  | 7                        |
| 12    | 62   | 29.00          | 29.00              | 51.00                  | 24                       |

Table 9 shows that January had the highest mean overall AQI at 36.82 and the largest number of poor air quality days with 35 days. March and December also had relatively high numbers of poor air quality days. July and August had the lowest mean AQI values, with July having no poor air quality days under the study threshold. This monthly pattern reinforces the seasonal findings and suggests that time based features may provide useful predictive value.

### Correlation Analysis

**Figure 6**

*Correlation Heatmap of Key Variables*

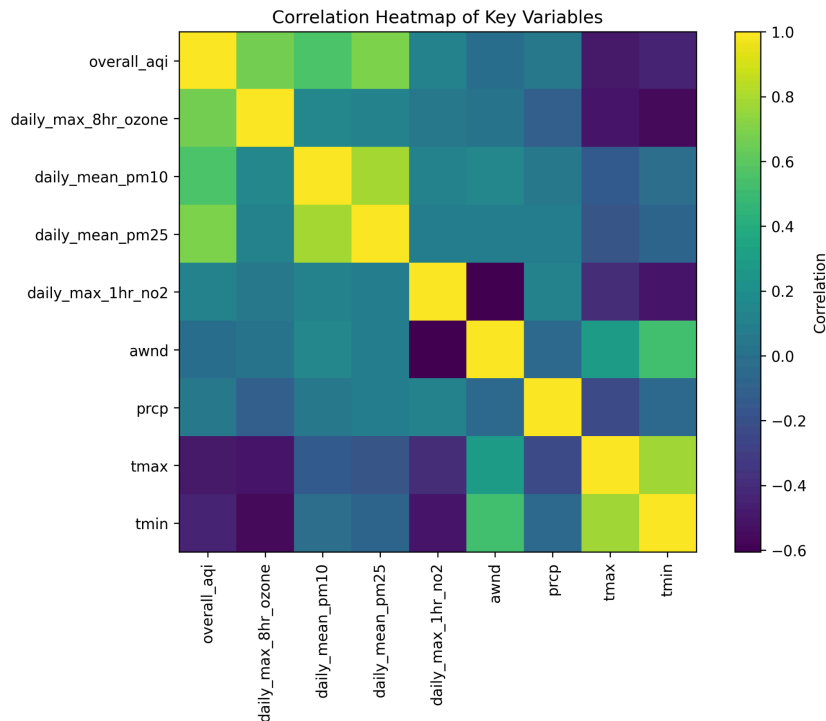


Figure 6 displays correlations among the main pollutant and meteorological variables.

The heatmap helps identify which predictors are most strongly related to overall AQI and

whether any variables may show overlapping information. This supports RQ1 and informs the modeling feature set.

**Table 10**

*Correlation of Key Predictors with Overall AQI*

| <b>Variable</b>        | <b>Correlation with Overall AQI</b> |
|------------------------|-------------------------------------|
| Daily mean PM2.5       | 0.686                               |
| Daily max 8 hour ozone | 0.670                               |
| Daily mean PM10        | 0.560                               |
| Daily max 1 hour NO2   | 0.134                               |
| Precipitation          | 0.035                               |
| Average wind speed     | -0.007                              |
| Minimum temperature    | -0.449                              |
| Maximum temperature    | -0.492                              |

Table 10 shows that PM2.5, ozone, and PM10 had the strongest positive correlations with overall AQI. PM2.5 had the highest correlation at 0.686, followed by ozone at 0.670 and PM10 at 0.560. NO2 had a weaker positive correlation. Among meteorological variables, maximum and minimum temperature had moderate negative correlations with overall AQI, while precipitation and wind speed had very weak relationships. These findings suggest that pollutant variables are more directly associated with overall AQI than the weather variables in this preliminary dataset.

**EDA Insight Summary****Table 11***EDA Insight Summary*

| <b>Figure/Table</b> |                        | <b>Why it Matters (Link to Decision/Next</b>                            |                                      |                                                       |
|---------------------|------------------------|-------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------|
| <b>Reference</b>    | <b>What it Shows</b>   | <b>Key Insight</b>                                                      | <b>RQ/objective)</b>                 | <b>Step</b>                                           |
| Figure 1            | Descriptive statistics | Overall AQI values were generally low to moderate, with a maximum of 56 | Supports target construction and RQ2 | Conduct threshold sensitivity review                  |
|                     |                        | `overall_aqi > 35` produced 150 poor air quality days                   |                                      | Use this target for preliminary classification        |
|                     |                        | Higher thresholds created severe class imbalance                        | Supports modeling feasibility        | Select `overall_aqi > 35` as study specific threshold |
|                     |                        | Overall AQI Elevated AQI                                                | Supports RQ3                         | Include temporal                                      |
| Table 6             | Target distribution    | Higher thresholds created severe class imbalance                        | Supports RQ2 and RQ4                 |                                                       |

|                  |                    |                          |                     |
|------------------|--------------------|--------------------------|---------------------|
| over time        | days appear to     |                          | features and        |
|                  | cluster in certain |                          | seasonal analysis   |
|                  | periods            |                          |                     |
|                  | Ozone and          |                          |                     |
|                  | PM2.5 show         |                          |                     |
| Pollutant        | stronger visible   |                          | Include pollutant   |
| specific AQI     | contribution to    |                          | variables in        |
| over time        | daily AQI          | Supports RQ1             | modeling            |
|                  | Ozone              |                          |                     |
|                  | dominated most     |                          | Interpret ozone and |
| Dominant         | days, followed     |                          | PM2.5 as key        |
| pollutant counts | by PM2.5           | Supports RQ1             | preliminary drivers |
|                  | Winter and         |                          |                     |
|                  | spring had the     |                          |                     |
|                  | highest mean       |                          |                     |
|                  | AQI and most       |                          | Conduct inferential |
| Seasonal AQI     | poor air quality   |                          | seasonal tests in   |
| patterns         | days               | Supports RQ3             | final report        |
| Monthly AQI      | January, March,    | Supports RQ3 and feature | Include `month` as  |
| patterns         | April, and         | engineering              | a model feature     |

|              |                  |              |                    |
|--------------|------------------|--------------|--------------------|
|              | December         |              |                    |
|              | showed higher    |              |                    |
|              | poor air quality |              |                    |
|              | counts           |              |                    |
|              | PM2.5, ozone,    |              |                    |
|              | and PM10 had     |              |                    |
|              | strongest        |              |                    |
|              | positive         |              |                    |
| Correlation  | correlations     |              | Compare with       |
| heatmap and  | with overall     |              | feature importance |
| correlations | AQI              | Supports RQ1 | from random forest |

### Summary of Analysis Findings

The analysis confirmed that the raw EPA and NOAA data could be cleaned and merged into a complete daily analytical dataset for the study period. The final cleaned dataset contains 731 daily observations, zero duplicate dates, and the necessary pollutant, meteorological, AQI, target, and time based variables. The selected target definition, `'poor_air_quality = 1'` when `'overall_aqi > 35'`, was chosen after reviewing threshold sensitivity and class imbalance.

The EDA results suggest that ozone and PM2.5 are especially important contributors to daily air quality patterns in this dataset. Seasonal and monthly summaries indicate that higher AQI values and poor air quality days were more common in winter and spring, especially

January and March. Correlation analysis showed that PM2.5, ozone, and PM10 had the strongest positive relationships with overall AQI, while temperature variables showed moderate negative relationships.

## **Modeling**

The modeling stage focused on the binary classification of poor air quality days using the cleaned daily analytical dataset. The target variable was `poor\_air\_quality`, which was derived from `overall\_aqi` using the study specific threshold of `overall\_aqi > 35`. Under this definition, the dataset contained 581 non-poor air quality days and 150 poor air quality days. This target distribution was suitable for preliminary classification, although the positive class remained smaller than the negative class.

## **Choice of Models with Justification**

Four candidate classification models were selected: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. These models were chosen because they provide a practical balance between interpretability, predictive performance, and feasibility for this capstone scale dataset. The goal was not only to identify the highest performing model, but also to compare models that differ in complexity and interpretability.

Logistic Regression was selected as the baseline model because it is widely used for binary classification and provides interpretable results. It estimates the probability that a given day belongs to the poor air quality class based on the selected pollutant, meteorological, and time

based predictors. Logistic Regression is useful for establishing a transparent benchmark against which more flexible models can be compared.

Decision Tree was selected because it can model nonlinear classification rules and is easy to interpret. A decision tree can split the dataset into decision paths based on pollutant and weather values, which makes it useful for explaining how certain conditions may lead to classification as a poor air quality day. However, a single decision tree can be sensitive to the training data, so its performance must be compared carefully against more stable models.

Random Forest was selected because it combines multiple decision trees to improve stability and reduce overfitting compared with a single decision tree. Random Forest is also useful because it provides feature importance values, which helps identify which variables contribute most strongly to classification. This makes Random Forest especially relevant to both RQ1, which examines important predictors, and RQ4, which compares model performance.

Support Vector Machine was selected because it can handle nonlinear classification boundaries through kernel based learning. This model was included to compare tree based methods with a margin based classifier. Support Vector Machine can perform well when classes are separated by complex boundaries, although it is generally less interpretable than Logistic Regression or Decision Tree.

Class weighting was used where appropriate because the target distribution was moderately imbalanced. The positive class represented 20.52% of the dataset, so model evaluation needed to consider more than accuracy alone. Class weighting helped reduce the risk that models would favor the majority class.

## Features Included and Feature Engineering

The preliminary feature set included pollutant concentration variables, meteorological variables, and selected time based features. Pollutant variables were included because they directly describe daily air quality conditions. Meteorological variables were included because weather conditions may influence pollutant accumulation, dispersion, or daily variation. Time based features were included because the EDA suggested seasonal and monthly variation in overall AQI and poor air quality days.

**Table 12**

*Feature set and usage*

| Feature                     | Original/Engineered   | Type                                 | Reason for Inclusion                                                                 | Used in Model(s) |
|-----------------------------|-----------------------|--------------------------------------|--------------------------------------------------------------------------------------|------------------|
| `daily_max_8h`<br>`r_ozone` | Original EPA variable | Continuous<br>pollutant<br>predictor | Ozone was frequently the dominant pollutant and strongly correlated with overall AQI | Yes              |
| `daily_mean_pm10`           | Original EPA variable | Continuous<br>pollutant<br>predictor | PM10 showed a positive correlation with overall AQI                                  | Yes              |
| `daily_mean_pm25`           | Original EPA variable | Continuous                           | PM2.5 was a major                                                                    | Yes              |

| <b>Feature</b> | <b>Original/Engineered</b> | <b>Type</b>    | <b>Reason for Inclusion</b>                                                         | <b>Used in Model(s)</b> |
|----------------|----------------------------|----------------|-------------------------------------------------------------------------------------|-------------------------|
| m25`           |                            | pollutant      | contributor to elevated                                                             |                         |
|                |                            | predictor      | AQI and showed the strongest correlation with overall AQI                           |                         |
| `daily_max_1h` | Original EPA variable      | Continuous     | NO2 was included as a monitored pollutant and                                       | Yes                     |
| r_no2`         |                            | pollutant      | potential air quality contributor                                                   |                         |
|                |                            | predictor      |                                                                                     |                         |
|                |                            |                |                                                                                     |                         |
| `tmax`         | Original NOAA variable     | Continuous     | Maximum temperature showed a moderate negative relationship with overall AQI in EDA | Yes                     |
|                |                            | meteorological |                                                                                     |                         |
|                |                            | predictor      |                                                                                     |                         |
|                |                            |                |                                                                                     |                         |
| `tmin`         | Original NOAA variable     | Continuous     | Minimum temperature was included to represent daily temperature conditions          | Yes                     |
|                |                            | meteorological |                                                                                     |                         |
|                |                            | predictor      |                                                                                     |                         |
|                |                            |                |                                                                                     |                         |
| `prcp`         | Original NOAA variable     | Continuous     | Precipitation may influence pollutant                                               | Yes                     |
|                |                            | meteorological |                                                                                     |                         |

| Feature      | Original/Engineered    | Type                      | Reason for Inclusion                                          | Used in Model(s) |
|--------------|------------------------|---------------------------|---------------------------------------------------------------|------------------|
| `awnd`       | Original NOAA variable | predictor                 | removal or dispersion                                         | Yes              |
|              |                        | Continuous meteorological | Average wind speed may influence pollutant                    |                  |
|              |                        | predictor                 | dispersion                                                    |                  |
| `month`      | Engineered from `date` | Time feature              | Monthly EDA showed variation in AQI and poor air quality days | Yes              |
|              |                        |                           | Included to test whether                                      |                  |
| `is_weekend` | Engineered from `date` | Binary time feature       | weekday/weekend timing contributes to classification          | Yes              |
|              |                        |                           |                                                               |                  |

Before modeling, missing numeric predictor values were handled using median imputation. Median imputation was selected because it is simple, transparent, and appropriate for a preliminary modeling stage where the goal is to retain all daily observations. The final preliminary modeling dataset contained 731 observations and 10 predictor variables.

The data were split into training and testing subsets using an 80/20 stratified split. Stratification preserved the proportion of poor air quality days in both the training and test sets.

The training set contained 584 observations, including 464 non-poor air quality days and 120 poor air quality days. The test set contained 147 observations, including 117 non-poor air quality days and 30 poor air quality days.

### Evaluation Metrics (Include Formulae and Calculations)

The models were evaluated using accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC-AUC). These metrics were selected because the project is a binary classification problem and because accuracy alone can be misleading when the target classes are imbalanced.

**Table 13**

*Evaluation metrics and formulae*

| Metric    | Formula                       | Interpretation                                                                         | Used For                           |
|-----------|-------------------------------|----------------------------------------------------------------------------------------|------------------------------------|
| Accuracy  | $= \frac{TP+TN}{TP+TN+FP+FN}$ | Proportion of all correct predictors                                                   | Classification                     |
| Precision | $= \frac{TP}{TP+FP}$          | Proportion of predicted poor air quality days that were actually poor air quality days | Overall classification correctness |
| Recall    | $= \frac{TP}{TP+FN}$          | Proportion of actual                                                                   | False-positive                     |

| Metric   | Formula                                                           | Interpretation                                                 | Used For                          |
|----------|-------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------|
|          |                                                                   | poor air quality days<br>correctly identified                  | control                           |
| F1-score | $F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$ | Balance between<br>precision and recall                        | Poor air quality<br>detection     |
|          | $TPR = \frac{TP}{TP + FN}$                                        |                                                                |                                   |
|          | $FPR = \frac{FP}{FP + TN}$                                        | Ability to distinguish<br>between classes across<br>thresholds | Overall<br>discrimination ability |
| ROC-AUC  | ROC-AUC = Area<br>under the ROC curve                             |                                                                |                                   |

In this project, recall and F1-score were especially important because the goal was not only to maximize total correct predictions, but also to identify poor air quality days reliably. Precision was also important because excessive false positives could reduce practical usefulness in a decision support context. ROC-AUC was included to compare model discrimination across thresholds rather than relying only on one classification cutoff.

### Validation Approach

Two validation approaches were used. First, the models were evaluated on the 20% holdout test set. This provided an initial assessment of performance on observations not used during model training. Second, stratified 5-fold cross-validation was conducted using the full

modeling dataset. Cross-validation was included to reduce dependence on a single train/test split and to provide a more stable estimate of model performance.

The same predictor set was used for all four models to ensure a fair comparison. The main comparison metrics were F1-score and ROC-AUC, with accuracy precision, and recall used as supporting measures. The model with the strongest balance of predictive performance and interpretability would be considered the best preliminary candidate for the final report development.

### Preliminary results

This section presents the preliminary findings from the cleaned dataset, exploratory analysis, and initial classification models. The results should be interpreted as interim findings because additional statistical testing, model refinement, and final interpretation will be completed for the final report.

#### Figure 7

*Preliminary Model F1 Comparison*

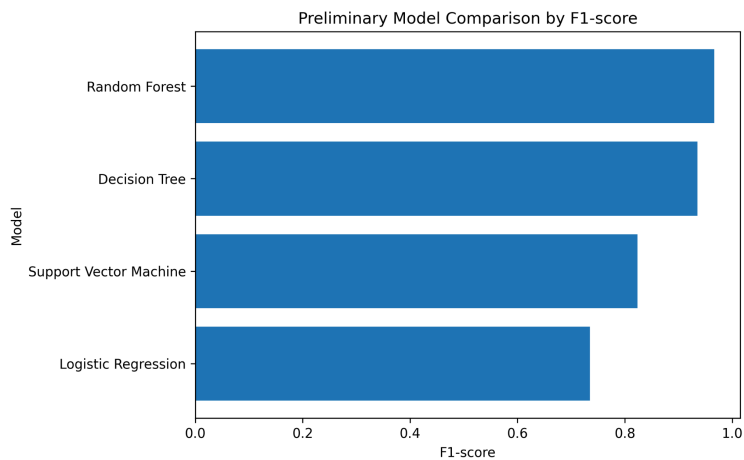


Figure 7 compares the four candidate models using F1-score from the preliminary train/test evaluation. Random Forest produced the highest F1-score, indicating the strongest balance between precision and recall for classifying poor air quality days.

### Figure 8

#### *Cross-Validated Model F1 Comparison*

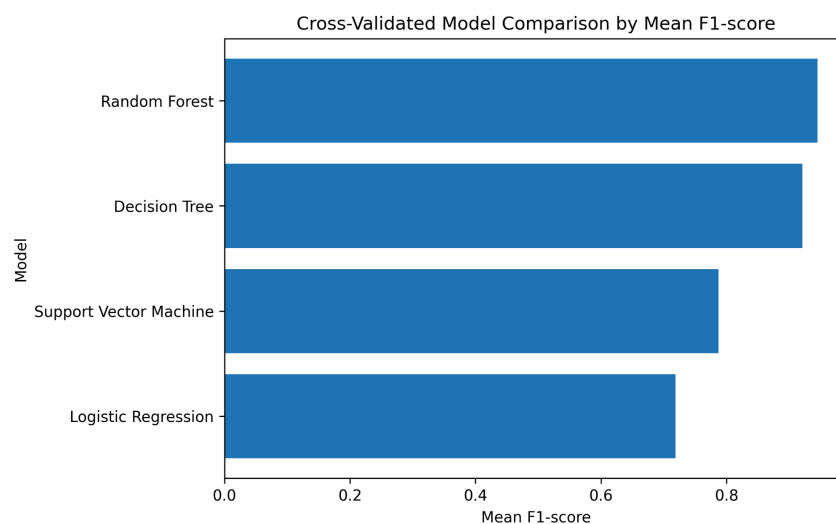


Figure 8 shows the stratified 5-fold cross-validation results by mean F1-score. Random Forest remained the strongest model across validation folds, which supports its stability beyond a single train/test split.

Figure 9

Random Forest Feature Importance

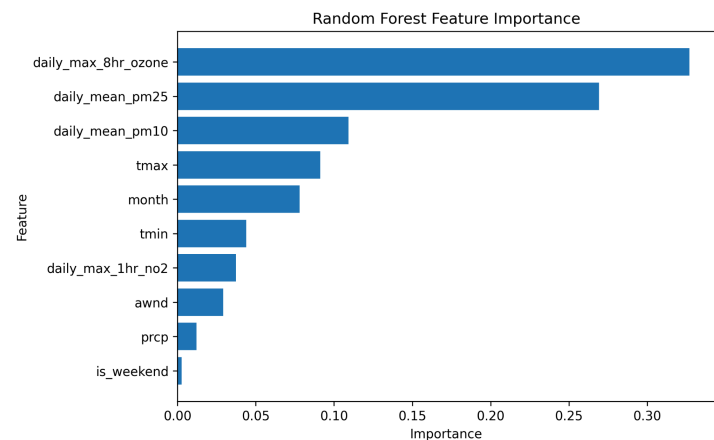


Figure 9 displays the most influential predictors in the Random Forest model. Ozone and PM2.5 had the highest feature importance values, suggesting that these pollutant variables contributed most strongly to the preliminary classification results.

Figure 10

Random Forest Confusion Matrix

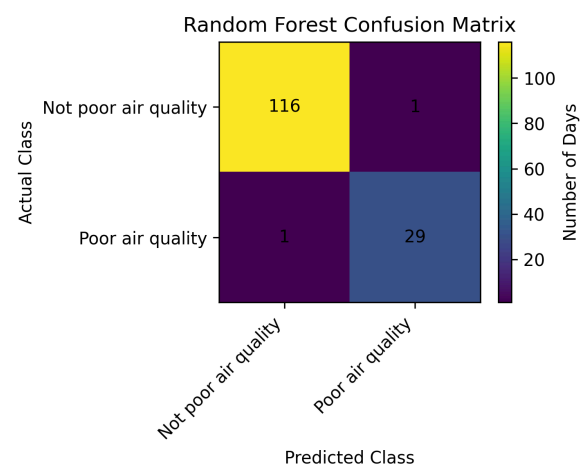


Figure 10 shows the Random Forest confusion matrix on the test set. The model correctly classified 116 non-poor air quality days and 29 poor air quality days, with one false positive and one false negative.

### **Preliminary Findings (by Research Question recommended)**

For RQ1, figure 9 shows that preliminary EDA results suggest that pollutant variables are more strongly associated with overall AQI than the meteorological variables. PM2.5 had the strongest positive correlation with overall AQI, followed by ozone and PM2.5. The Random Forest feature importance results also showed that ozone and PM2.5 were the two most important predictors in the preliminary classification model. These findings suggest that ozone and PM2.5 are the strongest preliminary drivers of poor air quality classification in the current dataset.

For RQ2, the preliminary model results suggest that poor air quality days can be classified with strong performance using the selected pollutant, meteorological, and time based predictors. The final binary target used the study specific threshold `overall\_aqi > 35`, which produced 150 poor air quality days and 581 non-poor air quality days. This target distribution was suitable for preliminary classification, although the positive class remained smaller than the negative class.

For RQ3, the descriptive results suggest seasonal and monthly variation in poor air quality days. Winter and spring had the highest mean overall AQI value and the largest number of poor air quality days. January had the highest monthly mean overall AQI and the largest number of poor air quality days. These patterns suggest that season and month may be relevant

temporal predictors. Formal inferential testing will be completed in the final report before drawing conclusions about statistical significance.

For RQ4, the Random Forest performed best among the four preliminary models. It achieved the strongest balance of accuracy, precision, recall F1-score, and ROC-AUC in both the train-test evaluation and the stratified 5-fold cross-validation results. Decision Tree was the second best model, followed by Support Vector Machine and Logistic Regression. These findings suggest that tree based methods are currently better suited to the structure of the dataset.

#### **Preliminary Model Performance (if modeling done)**

Four models were compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The Random Forest model performed best in the initial test set evaluation, with an accuracy of 0.986, precision of 0.967, recall of 0.967, F1-score of 0.967, and ROC-AUC of 0.994. The Random Forest confusion matrix showed 116 true negatives, 29 true positives, one false positive, and one false negative.

Stratified 5-fold cross-validation was also conducted to provide a more stable performance estimate. Random Forest remained the best performing model, with a cross-validated mean accuracy of 0.978, mean precision of 0.973, mean recall of 0.920, mean F1-score of 0.945, and mean ROC-AUC of 0.991. This supports Random Forest as the strongest preliminary model for the current dataset.

**Table 14***Preliminary Model Performance Comparison*

| <b>Model</b>           | <b>Validation Approach</b> | <b>Accuracy</b> | <b>Precision</b> | <b>Recall</b> | <b>F1-score</b> | <b>ROC-AUC</b> | <b>Key Takeaway</b>          |
|------------------------|----------------------------|-----------------|------------------|---------------|-----------------|----------------|------------------------------|
|                        | 5-f                        |                 |                  |               |                 |                |                              |
| Random Forest          | old cross-validation       | 0.978           | 0.973            | 0.920         | 0.945           | 0.991          | Best overall model           |
|                        | 5-f                        |                 |                  |               |                 |                |                              |
| Decision Tree          | old cross-validation       | 0.967           | 0.916            | 0.927         | 0.921           | 0.967          | Strong second best model     |
|                        | 5-f                        |                 |                  |               |                 |                |                              |
| Support Vector Machine | old cross-validation       | 0.899           | 0.698            | 0.907         | 0.787           | 0.971          | High recall, lower precision |
|                        | 5-f                        |                 |                  |               |                 |                |                              |
| Logistic Regression    | old cross-validation       | 0.858           | 0.611            | 0.880         | 0.719           | 0.941          | Useful baseline model        |

The preliminary results suggest that Random Forest provides the strongest classification performance while also supporting interpretation through feature importance. However, because the target variable is derived from `overall\_aqi`, and `overall\_aqi` is calculated from pollutant specific AQI values, the results should be interpreted as classification performance within the constructed analytical dataset rather than as final evidence of an operational forecasting system.

### **Interim Limitations and Risks**

- The threshold `overall\_aqi > 35` is a study specific operational definition of locally elevated AQI and is not an official EPA health category cutoff.
- Some pollutant values were missing in the raw EPA files and required median imputation for preliminary modeling.
- The NOAA weather station is a nearby meteorological station, but it is not located at the exact EPA Kapolei monitoring site.
- The current model results are preliminary and require further statistical testing, model refinement, and final validation before final conclusions are made.
- The high model performance should be interpreted carefully because `poor\_air\_quality` is derived from `overall\_aqi`, which is closely related to pollutant specific AQI measures.

### **Next Steps (for Final Report)**

- ☐ Conduct formal inferential testing for seasonal and temporal variation in poor air quality days.

- ☐ Refine interpretation of the best performing model and its feature importance results.
- ☐ Complete final validation of the selected model.
- ☐ Strengthen the discussion of practical application for local environmental monitoring and decision support.
- ☐ Expand the limitations section with final data and modeling constraints.
- ☐ Develop final recommendations based on the completed analysis.
- ☐ Present a complete interpretation of results by research question.

### Bibliography

- Eren, B., Serat, S., Damar Arifoglu, Y., & Ozdemir, S. (2025). Seasonal analysis and machine learning-based prediction of air pollutants in relation to meteorological parameters: A case study from Sakarya, Türkiye. *Applied Sciences*, 15(8), 4551. <https://doi.org/10.3390/app15084551>
- Islam, F., Nukala, S. K., Shrestha, P., Badgery-Parker, T., & Foo, F. (2025). Air pollution and cardiovascular disease: A systematic review of the effects of air pollution, including bushfire smoke, on cardiovascular disease. *American Heart Journal Plus: Cardiology Research and Practice*, 54, 100546. <https://doi.org/10.1016/j.ahjo.2025.100546>
- Kamińska, J. A. (2018). The use of random forests in modelling short-term air pollution effects based on traffic and meteorological conditions: A case study in Wrocław. *Journal of Environmental Management*, 217, 164-174. <https://doi.org/10.1016/j.jenvman.2018.03.094>
- Qi, L., Zheng, H., Ding, D., & Wang, S. (2024). A comparison of meteorological normalization of PM<sub>2.5</sub> by multiple linear regression, general additive model, and random forest methods. *Atmospheric Environment*, 338, 120854. <https://doi.org/10.1016/j.atmosenv.2024.120854>
- Sachdeva, S., Kaur, R., Kimmi, Singh, H., Aggarwal, K., et al. (2024). Meteorological AQI and pollutants concentration-based AQI predictor. *International Journal of Environmental Science and Technology*, 21, 4979-4996. <https://doi.org/10.1007/s13762-023-05307-8>

- Sigamani, S., & Venkatesan, R. (2022). Air quality index prediction with influence of meteorological parameters using machine learning model for IoT applications. *Arabian Journal of Geosciences*, 15, 340. <https://doi.org/10.1007/s12517-022-09578-2>
- Tırink, S. (2025). Machine learning-based forecasting of air quality index under long-term environmental patterns: A comparative approach with XGBoost, LightGBM, and SVM. *PLOS One*, 20(10), e0334252. <https://doi.org/10.1371/journal.pone.0334252>
- Wan Mahiyuddin, W. R., Ismail, R., Mohammad Sham, N., Ahmed, N. I., & Nik Hassan, N. M. N. (2023). Cardiovascular and respiratory health effects of fine particulate matters (PM<sub>2.5</sub>): A review on time series studies. *Atmosphere*, 14(5), 856. <https://doi.org/10.3390/atmos14050856>
- Wang, C., Liu, S., Yang, L., Wang, R., Tang, J., Zhang, W., & Wang, L. (2025). Time-series data-driven PM<sub>2.5</sub> forecasting: From theoretical framework to empirical analysis. *Atmosphere*, 16(3), 292. <https://doi.org/10.3390/atmos16030292>
- Zhou, S., Wang, W., Zhu, L., Qiao, Q., & Kang, Y. (2024). Deep-learning architecture for PM<sub>2.5</sub> concentration prediction: A review. *Environmental Science and Ecotechnology*, 17, 100400. <https://doi.org/10.1016/j.esec.2024.100400>